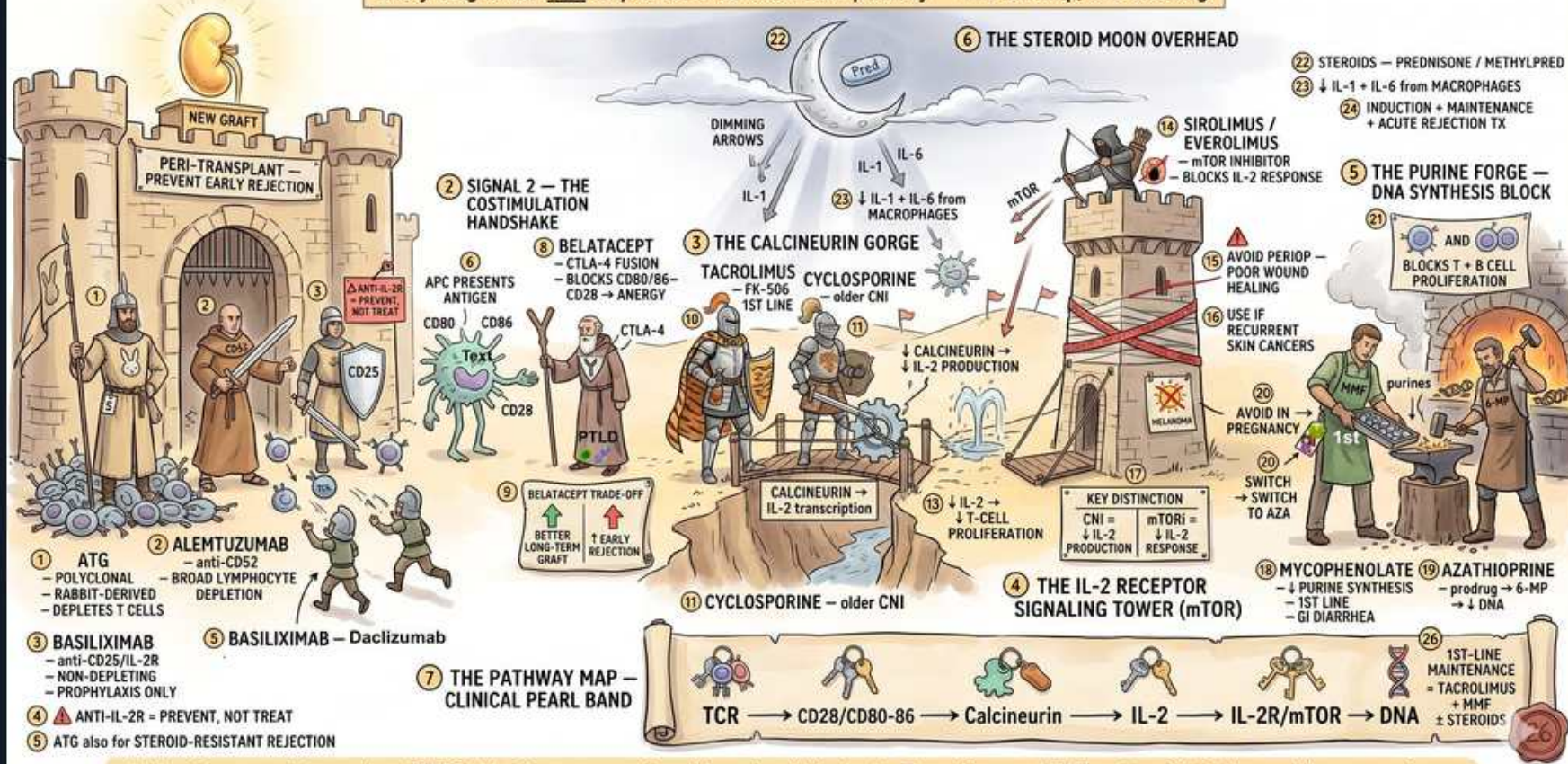


Kidney Transplant — Immunosuppression by Signal (Management 1)

THE T-CELL BATTLEFIELD — TRANSPLANT IMMUNOSUPPRESSION: INDUCTION GATE + MAINTENANCE BY SIGNAL

Every drug blocks **ONE** step of the T-cell activation pathway — find the step, find the drug



Induction = peri-transplant TRIO. Maintenance = signal-by-signal blockade. Tacrolimus + MMF $\pm$ steroids is the workhorse regimen.

1. Induction graft gate

WHAT YOU SEE

New graft at castle gate, 'peri-transplant — prevent early rejection'

WHAT IT ENCODES

Induction is intense immunosuppression given at the time of transplant (day 0, peri-operatively) to blunt the strong early alloimmune response and prevent acute rejection during the highest-risk first weeks. Agents are antibody-based: depleting (ATG, alemtuzumab) for high immunologic risk, or non-depleting (basiliximab) for standard/low risk. Induction also permits lower/delayed calcineurin inhibitor dosing early on. Mnemonic: induction = brief, intense 'gatekeeper'; maintenance = lifelong, lower-intensity defense.

BOARD TRAP

Board trap: confusing induction with maintenance. Induction is a one-time peri-transplant burst (basiliximab/ATG), NOT the daily lifelong regimen. A stem asking 'which drug does the patient take indefinitely to prevent chronic rejection' is maintenance (tacrolimus + MMF + steroids), not an induction antibody.

2. ATG (depleting)

WHAT YOU SEE

Polyclonal rabbit antibody icon, 'depletes T-cells'

WHAT IT ENCODES

Anti-thymocyte globulin (ATG, typically rabbit/Thymoglobulin) is a polyclonal antibody that profoundly DEPLETES circulating T cells; used as induction in high immunologic risk (high PRA, D/R sensitization, repeat transplant, delayed graft function) and as treatment for steroid-resistant acute cellular rejection. Side effects: cytokine-release syndrome (premedicate with steroids/antihistamine/acetaminophen), leukopenia/thrombocytopenia, serum sickness, and markedly increased CMV/EBV/PTLD risk requiring prophylaxis.

BOARD TRAP

Board trap: ATG-induced profound T-cell depletion is a classic CMV reactivation trigger — a febrile, leukopenic patient weeks after ATG for rejection is likely CMV, not 'expected ATG marrow effect' alone. Also, ATG (depleting) ≠ basiliximab (non-depleting); only the depleting agent is appropriate for established/steroid-resistant rejection.

3. Alemtuzumab

WHAT YOU SEE

Anti-CD52 icon, 'lymphocyte depletion'

WHAT IT ENCODES

Alemtuzumab is a monoclonal anti-CD52 antibody that causes prolonged depletion of T and B lymphocytes (and monocytes/NK cells); used as a potent depleting induction agent, sometimes allowing steroid- or CNI-minimization protocols. Because lymphocyte recovery is slow, it carries a sustained risk of opportunistic infection (CMV, PJP, fungal) mandating prolonged prophylaxis. CD52 is the target shared with its use in MS and CLL.

BOARD TRAP

Board trap: like ATG, alemtuzumab is depleting and immunosuppressively profound — answering that it is a 'mild, non-depleting' agent (the basiliximab role) is wrong. The prolonged lymphopenia means infection prophylaxis cannot be stopped early.

4. Basiliximab

WHAT YOU SEE

Anti-CD25 / IL-2R icon, 'non-depleting'

WHAT IT ENCODES

Basiliximab is a monoclonal antibody against CD25 (the IL-2 receptor alpha chain) that blocks IL-2-driven T-cell proliferation WITHOUT depleting T cells; it is the standard NON-depleting induction agent for low/standard immunologic risk recipients. It is very well tolerated (no cytokine-release syndrome, no added CMV/PTLD burden beyond baseline), but it is purely preventive and has no role in treating established rejection.

BOARD TRAP

Board trap: basiliximab does NOT deplete T cells and cannot reverse an established/ongoing acute rejection — choosing basiliximab to TREAT acute cellular rejection is wrong (answer is pulse steroids, then ATG if steroid-resistant). It is induction/prophylaxis only.

5. Signal 1 — TCR/antigen

WHAT YOU SEE

APC presenting antigen via CD80/86 to TCR

WHAT IT ENCODES

Signal 1 is antigen recognition: the antigen-presenting cell displays donor MHC/peptide to the T-cell receptor (TCR)-CD3 complex, initiating T-cell activation. Downstream this raises intracellular calcium, activating calcineurin → NFAT → IL-2 transcription. Calcineurin inhibitors (tacrolimus, cyclosporine) and steroids act on this proximal arm.

Mnemonic: 'Signal 1 = the antigen handshake at the TCR.'

BOARD TRAP

Board trap: don't confuse Signal 1 (TCR/antigen, blocked downstream by CNIs at calcineurin) with Signal 2 (costimulation, blocked by belatacept). Mislabeling which signal a drug blocks is the core mechanism distractor in these questions.

6. Signal 2 — costimulation

WHAT YOU SEE

Costimulation handshake CD28/CTLA-4; belatacept

WHAT IT ENCODES

Signal 2 is costimulation: T-cell CD28 binding APC B7 (CD80/CD86) is required for full activation; without it, the T cell becomes anergic. Belatacept is a CTLA-4–Ig fusion protein that outcompetes CD28 for B7, blocking costimulation; it is a CNI-sparing maintenance option that avoids nephrotoxicity but requires IV infusions. Mnemonic: 'Signal 2 = CD28/B7 handshake; belatacept = the CTLA-4 decoy.'

BOARD TRAP

Board trap: belatacept is CONTRAINDICATED in EBV-seronegative recipients because of a markedly increased risk of EBV-driven post-transplant lymphoproliferative disorder (PTLD), particularly CNS PTLD. Selecting belatacept for an EBV-naïve patient is a classic wrong answer; check EBV serostatus first.

7. Calcineurin gorge

WHAT YOU SEE

Knight in gorge, tacrolimus & cyclosporine block calcineurin

WHAT IT ENCODES

Calcineurin inhibitors (CNIs) bind immunophilins (tacrolimus→FKBP-12; cyclosporine→cyclophilin) to inhibit calcineurin, preventing NFAT dephosphorylation and thus blocking IL-2 transcription — the backbone of maintenance therapy. Tacrolimus is first-line (more potent, lower acute rejection). Both are nephrotoxic (afferent arteriolar vasoconstriction) and need therapeutic drug monitoring; both are metabolized by CYP3A4 (azoles, diltiazem raise levels; rifampin lowers them).

BOARD TRAP

Board trap: tacrolimus (FK-506) is preferred over cyclosporine as first-line CNI — choosing cyclosporine as the modern default is wrong. Distinguish their toxicities: tacrolimus → neurotoxicity (tremor), new-onset diabetes, alopecia; cyclosporine → gingival hyperplasia, hirsutism, hyperlipidemia. A rising Cr with a high tac level = CNI nephrotoxicity (reduce dose), NOT rejection (don't escalate).

8. IL-2 receptor / mTOR tower

WHAT YOU SEE

Signaling tower, IL-2 → IL-2R → mTOR

WHAT IT ENCODES

Once IL-2 is produced (downstream of Signal 1/calcineurin), it binds the IL-2 receptor (CD25) and activates the PI3K/mTOR pathway that drives clonal T-cell proliferation — 'Signal 3.' This axis is the target of two distinct drug classes: basiliximab blocks the IL-2 receptor (CD25) extracellularly, while sirolimus/everolimus block mTOR intracellularly. Mnemonic: IL-2 = the growth command, mTOR = the proliferation engine.

BOARD TRAP

Board trap: don't conflate the two ways to hit this axis. Basiliximab blocks the IL-2 RECEPTOR (induction, non-depleting); sirolimus blocks mTOR (the downstream proliferation signal). Both touch IL-2 signaling but at different points and for different roles.

9. Sirolimus / everolimus

WHAT YOU SEE

mTOR inhibitor blocks IL-2 response

WHAT IT ENCODES

mTOR inhibitors (sirolimus, everolimus) bind FKBP-12 and inhibit mTOR, blocking IL-2-driven T-cell proliferation; they are CNI-sparing and notably antiproliferative/antineoplastic (reduce skin cancer and Kaposi sarcoma). Characteristic side effects: impaired wound healing, mouth ulcers, proteinuria, hyperlipidemia, pneumonitis, and cytopenias. They are favored when malignancy (e.g., recurrent skin cancer) is a concern.

BOARD TRAP

Board trap: AVOID mTOR inhibitors in the immediate peri-operative period — impaired wound healing causes dehiscence and lymphocele; substituting sirolimus on day 0 is wrong (use a CNI-based regimen, switch later). Also, sirolimus is a recognized cause of new/worsening proteinuria — don't reflexively call that proteinuria 'rejection.'

10. Purine forge — DNA block

WHAT YOU SEE

Blacksmith blocking DNA / B-cell proliferation

WHAT IT ENCODES

Antimetabolites inhibit purine/DNA synthesis, suppressing proliferation of both T and B lymphocytes (which depend on de novo purine synthesis). This is the third pillar of triple maintenance therapy alongside a CNI and steroids. The two drugs are mycophenolate (inhibits IMPDH, the rate-limiting de novo purine enzyme) and azathioprine (a purine analog activated to thioguanine nucleotides).

BOARD TRAP

Board trap: antimetabolites are maintenance agents that suppress proliferation, not rapid-acting rescue drugs — they do not treat acute rejection (that's pulse steroids/ATG). Their main acute issue is myelosuppression, so a pancytopenic transplant patient should prompt review of MMF/azathioprine dosing and interactions.

11. Mycophenolate

WHAT YOU SEE

Anti-purine icon, 'GI diarrhea', 1st-line maintenance

WHAT IT ENCODES

Mycophenolate mofetil (MMF) is hydrolyzed to mycophenolic acid, a reversible inhibitor of inosine monophosphate dehydrogenase (IMPDH), selectively blocking de novo purine synthesis on which lymphocytes rely — it is the FIRST-LINE maintenance antimetabolite. Side effects: dose-limiting GI toxicity (diarrhea, nausea — enteric-coated form helps) and myelosuppression (leukopenia). No TPMT issue and no allopurinol interaction (unlike azathioprine).

BOARD TRAP

Board trap: MMF is TERATOGENIC (first-trimester loss, ear/facial malformations) and must be stopped and switched (e.g., to azathioprine) before conception — continuing MMF in a patient planning pregnancy is wrong. Also, new diarrhea in a transplant patient is often MMF, but CMV colitis must be excluded before simply attributing it to the drug.

12. Azathioprine

WHAT YOU SEE

Older antimetabolite icon

WHAT IT ENCODES

Azathioprine is an older antimetabolite, a prodrug converted to 6-mercaptopurine and then thioguanine nucleotides that incorporate into DNA; inactivation is via thiopurine methyltransferase (TPMT) and xanthine oxidase. Low/absent TPMT activity → severe myelosuppression (check TPMT before use). It is the preferred antimetabolite in pregnancy (safer than MMF). Main toxicities: dose-dependent myelosuppression and hepatotoxicity.

BOARD TRAP

Board trap: allopurinol (xanthine oxidase inhibitor) blocks azathioprine catabolism → toxic accumulation and life-threatening pancytopenia; giving full-dose azathioprine with allopurinol (or febuxostat) is a classic fatal-interaction wrong answer — reduce azathioprine ~75% or switch agents. Also remember azathioprine, not MMF, is the antimetabolite of choice in pregnancy.

13. Steroids moon

WHAT YOU SEE

Steroid moon overhead, prednisone/methylprednisolone

WHAT IT ENCODES

Corticosteroids (prednisone for maintenance, IV methylprednisolone for pulses) are broad immunosuppressants that inhibit NF- κ B, reducing macrophage/T-cell cytokine production (IL-1, IL-2, IL-6, TNF) and lymphocyte trafficking. They serve in all three roles: induction adjunct, low-dose maintenance, and FIRST-LINE pulse therapy for acute cellular rejection. Chronic toxicities: hyperglycemia, osteoporosis/avascular necrosis, infection, weight gain, cataracts, adrenal suppression.

BOARD TRAP

Board trap: high-dose pulse IV methylprednisolone is first-line for acute CELLULAR (T-cell-mediated) rejection — but steroids alone are inadequate for antibody-mediated rejection (needs plasmapheresis/IVIG $\pm$ rituximab). Don't treat a C4d+/DSA+ ABMR with steroid pulses alone.

14. Workhorse regimen band

WHAT YOU SEE

Boards band: TCR $\rightarrow$ CD80/86 $\rightarrow$ calcineurin $\rightarrow$ IL-2 $\rightarrow$ mTOR $\rightarrow$ DNA

WHAT IT ENCODES

Standard long-term maintenance is TRIPLE therapy targeting different points of the activation cascade: a calcineurin inhibitor (tacrolimus, Signal 1/IL-2 transcription) + an antimetabolite (mycophenolate, DNA/purine synthesis) + low-dose steroid (broad cytokine suppression). Layering mechanisms maximizes efficacy while limiting any single drug's toxicity. Reflex regimen for the boards: 'tacrolimus + MMF + prednisone.' Don't forget concurrent prophylaxis (TMP-SMX for PJP, valganciclovir for CMV).

BOARD TRAP

Board trap: the default maintenance answer is tacrolimus + mycophenolate + prednisone — substituting an induction antibody (basiliximab/ATG) into the lifelong regimen, or naming cyclosporine as first-line CNI, is wrong. Equally, forgetting that maintenance patients need PJP (TMP-SMX) and CMV prophylaxis is a common omission.